



PRACTICE EXAMINATION

Mathematical Methods

2026

Question booklet 2

- Questions 7 to 11 (50 marks)
- Answer **all** questions
- Write your answers in this question booklet
- You may write on page 14 if you need more space
- Allow approximately 65 minutes
- Approved calculators may be used — complete the box below

Examination information

Materials

- Question booklet 1
- Question booklet 2
- Formula sheet
- Student name label

Instructions

- Show appropriate working and steps of logic in the question booklets
- State all answers correct to three significant figures, unless otherwise instructed
- Use black or blue pen
- You may use a sharp dark pencil for diagrams and graphical representations

Total time: 130 minutes

Total marks: 100 (across both question booklets)

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Copy the information from your student name label here

Graphics calculator

1. Brand _____
Model _____
2. Brand _____
Model _____

Consider the function $f(x) = \frac{8}{x+1}$, where $x \geq 0$. The graph of $y = f(x)$ is shown in Figure 1, along with a shaded region. The shaded region is bounded by $y = f(x)$, the x -axis, the y -axis, and the vertical line $x = 4$.



(a) On Figure 1, draw four rectangles of equal width that could be used to calculate L_4 .

(b) Calculate the value of L_4 .

(c) Explain why L_4 is an *underestimate* of the area of the shaded region.

(2 marks)

(d) Using an algebraic approach, determine the *exact* area of the shaded region.



(2 marks)

(e) Determine the *exact* value of b , where $b > 0$, such that $\int_0^b f(x) \, dx = 20$.



(2 marks)

Consider the function $f(x) = x^2 \ln x$, where $x > 0$. The graph of $y = f(x)$ is shown in Figure 2.

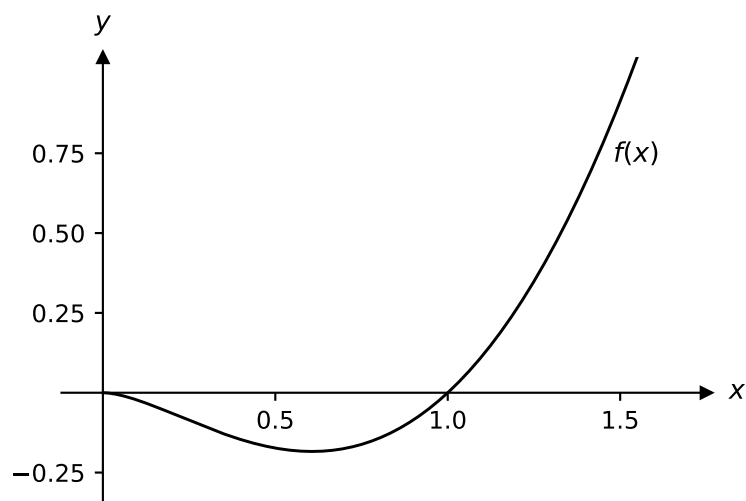


Figure 2

- (a) Show that $f'(x) = x(2\ln x + 1)$.

(2 marks)

- (b) Using an algebraic approach, determine the *exact* coordinates of the stationary point of $y = f(x)$.

(3 marks)

Question 9 (10 marks)

The length of time, in minutes, that a customer spends on a call to a technical help desk can be modelled by the continuous random variable X with probability density function

$$f(x) = kx(8 - x)$$

where $0 \leq x \leq 8$ and k is a positive constant. The graph of $y = f(x)$ is shown in Figure 3.

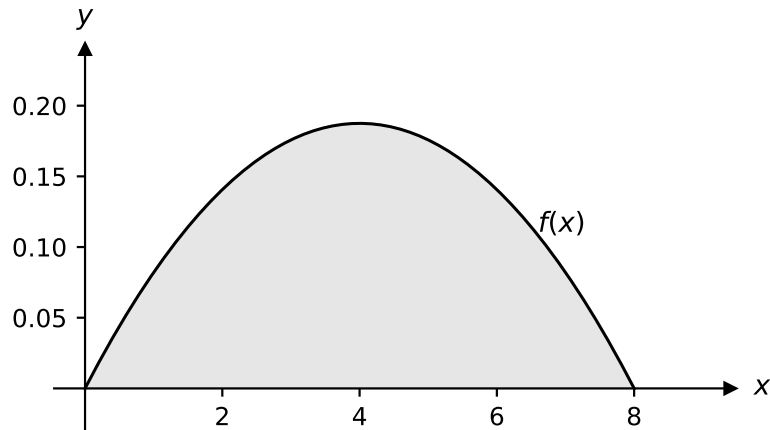


Figure 3

- (a) Use an algebraic approach to show that $k = \frac{3}{256}$.

(3 marks)

- (b) Determine $\Pr(X \leq 3)$.

(1 mark)

(d) The standard deviation of X is $\sigma_X = 1.789$ minutes (correct to four significant figures).

(i) Calculate a 95% confidence interval for the mean length of calls to the help desk.

(ii) The manager claims that the mean length of calls has *increased* from 4 minutes. Does the confidence interval calculated in part (d)(i) support this claim with 95% confidence? Justify your answer.

PLEASE TURN OVER

The depth of the water at a jetty, D metres, t hours after midnight on a particular day can be modelled by the function

for $0 \leq t \leq 24$. The graph of $y = D(t)$ is shown in Figure 4.



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For each positive integer n , consider the function

Figure 5 shows the graphs of $y = f_1(x)$, $y = f_2(x)$, and $y = f_3(x)$. For $x > 0$, each of these functions has exactly one maximum turning point.



The x -coordinate of the maximum turning point of $y = f_n(x)$ for $x > 0$ is shown in Table 1 for $n = 1$, $n = 2$, and $n = 3$.

n	1	2	3
$f_n(x)$	xe^{-x}	x^2e^{-x}	x^3e^{-x}
x-coordinate of the maximum turning point	1	2	3

- (ii) Determine the smallest positive integer n for which the maximum value of $f_n(x)$ is greater than 100. Give evidence to support your answer.

(3 marks)

